



Alberta Public Health Disease Management Guidelines

Typhoid Fever



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness.^(A)

- Isolation of *Salmonella typhi* from an appropriate clinical specimen (e.g., feces, blood, urine, bone marrow).^(B)

Probable Case

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case.

Carrier

Individuals who continue to shed *S. typhi* for one year or greater are considered to be carriers.^(C)

^(A) Clinical illness is characterized by insidious onset of sustained fever, headache, malaise, anorexia, relative splenomegaly, constipation or diarrhea, and non-productive cough. Relative bradycardia and rose spots (less than 25% of individuals) may be seen. Atypical presentations occur and severity of illness varies. Chronic carrier state (< 5% of population) is usually linked to the biliary or urinary tract and should be distinguished from short-term faecal carriage.

^(B) Refer to the [Public Health Laboratories \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

^(C) Alberta Health maintains a Typhoid/Paratyphoid Registry for purposes of monitoring carriers as they potentially pose a long term health risk for transmission of disease.

Reporting Requirements

Physicians/Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results:

- by FMP to the MOH (or designate) of the zone, and
- by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and probable cases.
- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Typhoid fever is caused by *Salmonella enterica subsp. enterica serovar typhi* (commonly known as *S. typhi*).⁽¹⁾

Clinical Presentation

Typhoid is a systemic bacterial disease characterized by insidious onset of sustained fever, severe headaches, malaise, anorexia, a non-productive cough in the early stage of the illness, a relative bradycardia, splenomegaly and abdominal discomfort.⁽¹⁾ A transient, macular rash of rose-colored spots can occasionally be seen on the trunk.⁽²⁾ In adults, constipation is more often seen than diarrhea. The clinical picture can vary from mild illness to severe clinical disease with abdominal discomfort and other complications. The severity of illness is dependent on the infecting dose, the virulence of the bacterial strain, duration of the illness before initiation of appropriate treatment, age, and vaccine history.⁽¹⁾ Complications such as ulceration of Peyer's patches in the ileum causing intestinal haemorrhage or perforation (in about 1% of cases) can occur, especially late in untreated cases. Case fatality rate can fall to < 1% with prompt, appropriate antimicrobial treatment.⁽¹⁾ Relapses (generally milder than the initial clinical illness) can occur, depending on what antimicrobials are used in treatment.⁽¹⁾

Diagnosis

The causative organisms can be isolated from the blood early in the disease, and from urine and feces after the first week. Bone marrow culture provides the greatest sensitivity for bacteriological confirmation, even in persons who have already received antibiotics.⁽¹⁾

Treatment

- Generally fluoroquinolones are effective for treatment in adults, however, antibiotic resistance has become a public health problem in many areas (especially Asia) and therefore, clinicians should consider local resistance patterns when choosing appropriate antimicrobials for treatment. ^(1,9,10)
- Consultation with an infectious diseases specialist may be appropriate.

Reservoir

Humans are the sole reservoirs for typhoid. Family contacts may be transient or permanent carriers and inadvertently spread infection.⁽¹⁾

Transmission

Typhoid is transmitted by the fecal-oral route through ingestion of food and water contaminated by urine or feces from infected cases or carriers. The infection is rarely spread by casual contact. Shellfish (particularly oysters) taken from sewage-contaminated beds, raw fruits, vegetables fertilized by night soil (human excrement) and eaten raw, contaminated milk and milk products (usually contaminated by hands of carriers), are important sources of infection to consider. Flies may also infect foods; therein the organism can multiply to achieve an infective dose. The infective dose for typhoid is much lower than that of paratyphoid.⁽¹⁾

Incubation Period

The incubation period is usually 8–14 days but can range from three to more than 60 days, depending on the size of infecting dose and host factors.⁽¹⁾

Period of Communicability

The period of communicability lasts as long as the bacilli are present in the excreta. This usually begins from the first week after onset of illness and continues through convalescence and for a variable period thereafter. It is estimated that approximately 10% of untreated typhoid cases can continue to discharge bacilli for three months after onset of symptoms.⁽¹⁾

The carrier state may follow acute illness, mild or even subclinical infections. Between 2–5% of cases infected with *S. typhi* may become chronic carriers and can be more commonly seen in individuals (especially women) infected during middle age, and those who have biliary tract abnormalities, including gallstones. Chronic urinary carriers may occur with schistosome infections or kidney stones.⁽¹⁾

Host Susceptibility

Susceptibility is general and is increased in persons with achlorhydria (a condition in which production of gastric acid in the stomach is absent or low or more commonly, treatment with acid suppression agents) (S. Houston, 2013, Personal Communication, January 28). Relative specific immunity follows clinical disease, asymptomatic infection and immunization.⁽¹⁾

Incidence

General

Worldwide occurrence. Typhoid fever continues to be a public health concern in many developing countries, with school age children (aged 5–15) being disproportionately affected. In some endemic countries, incident rates in younger children (< 5 years) have been seen to be similar or exceed those of school age children. It is conservatively estimated by the World Health Organization that the annual global occurrence of typhoid is approximately 21 million cases, with 1–4% ending in death.⁽³⁾

In developed countries, where standards of sanitation are high, disease incidence is sporadic and mainly associated with foreign travel to endemic countries.⁽¹⁾

Canada

The incidence of typhoid is very low in Canada. From 2009–2012, the number of cases reported nationally ranged between 136 (2012) and 183 (2011) with a rate of 0.39–0.53 cases per 100,000 respectively.⁽⁴⁾ The greatest risk of typhoid infection for Canadians occurs while travelling to countries where sanitation is likely to be poor and associated with exposures to food and water in uncontrolled settings. (i.e., market stalls, street vendors, family settings).⁽⁵⁾ The risk of typhoid fever is highest for travellers to southern Asia.⁽²⁾

Alberta

The greatest risk of typhoid infection for Albertans appears to be travel related to endemic countries. Between 2000 and 2012, the number of cases of typhoid reported annually varied from 5 (2001) to 20 (2010) representing rates of 0.30 cases per 100,000 to 0.38 cases per 100,000 respectively. During this same period of time, 140/165 cases reported travel outside of Canada (e.g., travelled to South Asia), an additional 12/165 cases reported recent immigration to Canada and for 11/165 cases there is no information available. One of the 165 cases was epi-linked to a confirmed case that travelled outside of Canada.⁽⁶⁾

Public Health Management

Key Investigation

- Confirm the diagnosis.
- Obtain a history of illness including the date of onset and signs and symptoms.
- Identify any underlying medical conditions (e.g., decreased gastric acidity).
- Determine immunization history.
- Determine the occupation of the case (e.g., food handler, childcare facility worker, healthcare worker,) and identify specific duties at work.
- If the case is a child, determine attendance at a childcare facility (e.g., daycare, dayhome) or other childcare site.
- Determine whether the case had any other type of institutional contact (e.g., long term care).
- Determine the possible source of infection taking into account the incubation period, reservoir, and mode of transmission. Assessment should include the following.
 - Determine history of travel. **NOTE:** Additional specimens (e.g., urine for *S.typhi*/*S.paratyphi* +/- serology for schistosomiasis) may be indicated in special circumstances. Refer to [Management of a Case](#).
 - Determine history of residing in areas with poor sanitation, including improper water treatment and sewage disposal (including recent immigration).
 - Obtain a detailed food history (including consumption of shellfish).
 - Identify any risk behaviours including lifestyle risks for infection (e.g., high risk sexual practices especially contact with feces).
 - Determine any exposure (either household or non-household) to a confirmed case of typhoid.
 - Identify symptomatic household or other close contacts (e.g., travel companions or others) that have recently travelled to a developing country.
- Identify contacts that may have had exposure during the period that the case was infectious, especially those that are in sensitive situations or occupations (see Table 1). Consider the following individuals when identifying contacts:
 - household contacts,
 - recent co-travellers (includes anyone who travelled with the case and is likely to have been exposed to the same source of infection as the case rather than someone who only travelled on the same plane/bus as the case), or
 - other close, non-household contacts (e.g., sexual contacts, childcare site contacts).

Table 1: Sensitive Situations or Occupations (SSO)

SSO	Definition
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Healthcare, child care or other staff	• Has contact through serving food to highly susceptible persons. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a childcare facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene) and is involved in an activity that may promote disease transmission.

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

Management of a Case

General

- Provide information about disease transmission and the appropriate infection prevention and control measures to be implemented to minimize the possibility of transmission including strict hand hygiene especially:
 - after using the washroom,
 - after changing diapers, and
 - before preparing/handling and serving food.
- Advise the case to refrain from preparing food for others for the duration of the period of communicability.
- Advise all other cases (i.e., those not involved in SSO) to remain off work until they are free from diarrhea and other gastrointestinal symptoms and for 48 hours following resumption of normal stool.
 - Continued public health surveillance is not required for these cases; however, follow-up with the personal physician for clearance of the organism is recommended. Persons not involved in high risk activities/occupations present a minimal risk of spreading gastrointestinal pathogens if they are healthy and have normal, well-formed stools.

Hospitalized Patients

- Routine practices should be followed.
- For hospitalized children or adults with poor hygiene habits or who have incontinence that cannot be contained, additional precautions (i.e., contact precautions) should be implemented.⁽⁷⁾

Exclusion – SSO

- Exclude symptomatic and asymptomatic cases who are involved in SSO (see [Table 1](#)).
- Exclusion applies until two consecutive stool specimens collected from a confirmed case are reported as negative **AND** one urine culture for *S. typhi* is reported negative if case EVER travelled to or lived in a schistosomiasis endemic country and may have been exposed to schistosomiasis. Refer to the [Centers for Disease Control and Prevention website](#)⁽⁸⁾ and the algorithm in [Appendix 1](#) for more information.
 - Collect stool specimens after stools have returned to normal and at least 21 days following completion of antibiotic and not less than 24 hours apart.
 - If one or both stool samples are positive for *S. typhi*, continue submitting samples for testing at the specified intervals as outlined in [Appendix 1](#).
 - If urine sample is positive for *S. typhi*, collect serology for schistosomiasis:
 - eight weeks post exposure to areas where risk exists for infection with *Schistosoma japonicum*, *S. mansoni*, *S. mekongi* and *S. intercalatum* exists, or
 - twelve weeks post exposure to areas where risk exists for infection with *S. haematobium*.

Specimens collected before this time, may yield false negative results (Kowaleska-Grochowska, K., 2012, Personal Communication, November 19). See [Appendix 1](#).

 - If urine sample is positive for *S. typhi* and schistosomiasis, refer case back to physician and advise physician to treat case concurrently for both infections, even if this means repeating antibiotic treatment.
 - Persons co-infected with schistosomiasis should be treated with praziquantel to eliminate possible carriage of salmonella bacteria (including *S. typhi*) by the schistosomes.⁽¹⁾
- If possible, consideration should be given to temporary redeployment away from activities that involve increased risk of transmission.

Management of Contacts

- “Warn and inform” contacts by providing information about disease transmission and appropriate infection prevention and control measures. Stress the measures that need to be taken to minimize possible fecal-oral transmission including strict hand hygiene – especially after using the washroom, changing diapers, and before eating and preparing/handling foods.
- Refer all symptomatic contacts to their physician for assessment.
- Refer asymptomatic co-travellers who are not involved in SSO to their own health care provider for screening and follow-up.
- Public health follow-up is generally not required for contacts of a case who are not involved in SSO; however, the circumstances should be considered individually. Follow-up with their health care provider is appropriate.
- Additional contact follow-up and assessment of a source of infection may be necessary if cases are identified with no recent travel history and/or for whom travel is an unlikely source of infection. Stool samples may be requested on identified contacts to determine the source of the infection in the case.

Exclusions – SSO

- See [Appendix 2](#) for information on stool collection for contacts involved in SSO.
- Exclude **symptomatic contacts** involved in SSO until two consecutive stool specimens are taken not less than 24 hours apart and are reported as negative prior to returning to work or back to their childcare facility.
 - If any stool sample is positive, manage as a confirmed case.
 - If stool sample is negative, advise contact to:
 - remain home while symptomatic and until 48 hours after last symptoms before returning to work or childcare facility,
 - continue to self-monitor for symptoms within 28 days after initial negative stools, and
 - seek prompt medical attention and inform public health if symptoms develop.
- Screen **asymptomatic co-travellers** who are involved in SSO:
 - Collect one stool specimen and question about symptoms both while away travelling and since returning.
 - No exclusion is necessary for asymptomatic co-travellers while waiting for results of screening stool specimen. If stool result is positive, they would then be treated and managed as a confirmed case.
- Collect ONE screening stool for **asymptomatic household or other close, non-household contacts (e.g., sexual contacts)** involved in SSO and **asymptomatic childcare site contacts (e.g., staff and children in the same room)**.
 - Normally, exclusion is **not necessary** while waiting for results of screening.
 - If the individual is unable/unwilling to comply with good hand hygiene OR unwilling to provide a screening stool sample, exclusion is **necessary** prior to returning to work or back to their childcare facility and would apply until one stool specimen is submitted and reported as negative.
 - Consultation with the MOH is appropriate.
 - Contacts should be advised to self-monitor and seek prompt medical attention AND contact public health if they develop symptoms within 28 days of the initial negative stool result.

Management of Chronic Carriers and Their Contacts

Individuals who continue to shed *S. typhi* for one year or greater are considered **carriers**.

- Refer to [Appendix 1](#) for details on continued stool testing for carriers.
- Medical intervention may be indicated for individuals who become carriers, and consultation with an infectious diseases specialist should be considered.
- Exclude chronic carriers from activities that are involved in SSO and continue submitting stool specimens as outlined in [Appendix 1](#).
 - If possible, consideration may be given to temporary redeployment of the carrier away from activities that involve increased risk of transmission.

- **Contacts of carriers** should be advised to seek prompt medical assessment and notify public health if they become symptomatic.
 - Exclude symptomatic contacts of carriers who are involved in SSO until two consecutive stool specimens taken not less than 24 hours apart are reported as negative prior to returning to work or back to their childcare facility.
 - Asymptomatic contacts of carriers are not excluded and no stool specimens are required.
 - Public health follow up is generally not required for contacts of a carrier who are not involved in SSO; however, stool specimens may be requested to determine the source of the infection in the case.
 - Typhoid vaccine may be recommended, in consultation with the CMOH (or designate), for people with ongoing household or intimate exposure to *S. typhi* carriers. ^(5,11)

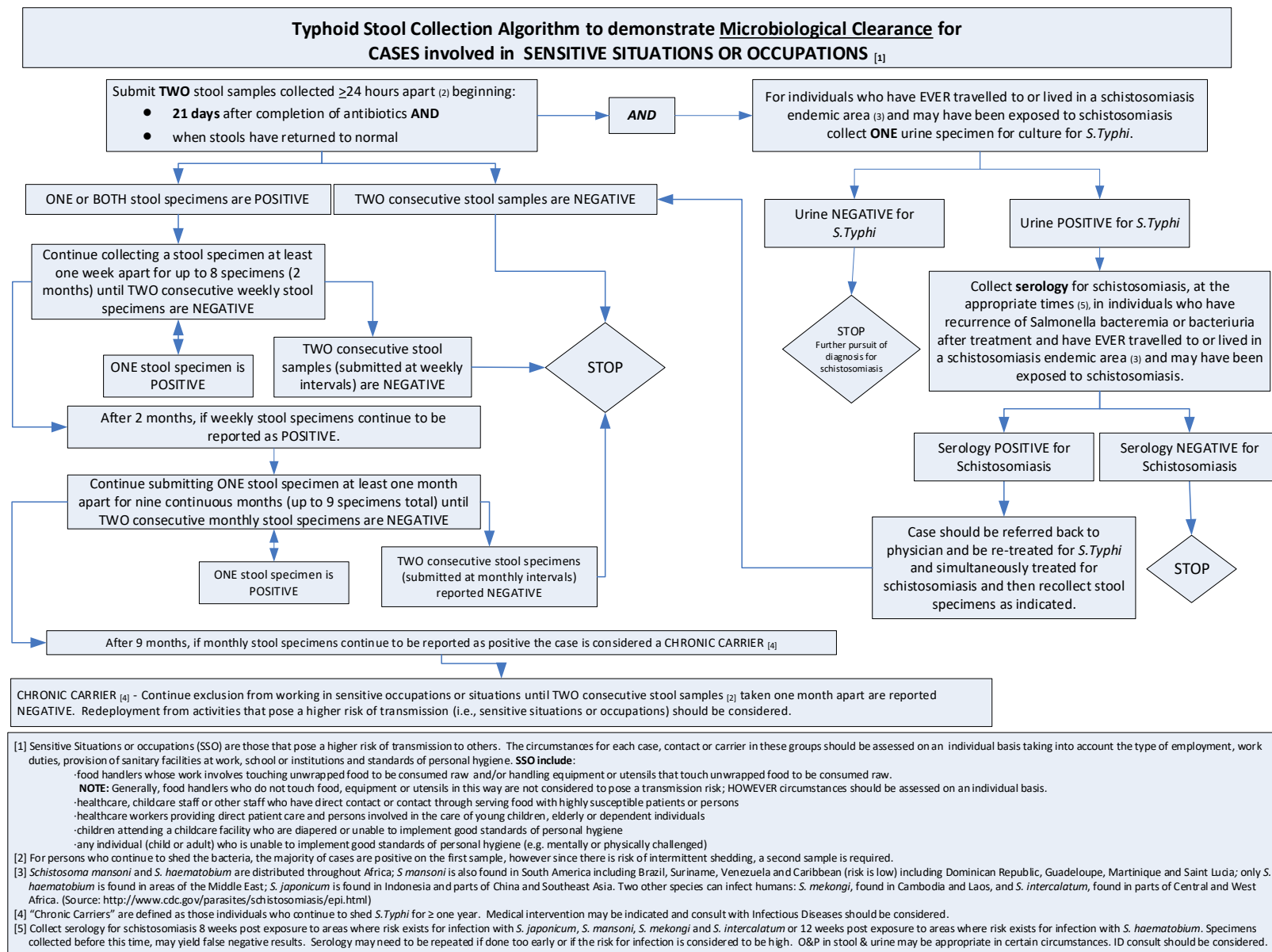
Preventive Measures

- Educate the public regarding the need for:
 - thorough hand washing, especially after using the washroom, changing diapers, and before eating and preparing/handling foods,
 - sanitary disposal of feces, and
 - hygienic food preparation and food/equipment handling practices.
- Reinforce information about boiling or steaming shellfish for at least 10 minutes before serving.
- Recommend typhoid vaccine for:
 - household or intimate contacts of a typhoid carrier (refer to the current [Alberta Immunization Policy](#) (AIP)).⁽¹¹⁾
 - laboratory workers who frequently handle cultures of *S. typhi* (refer to the current AIP).⁽¹¹⁾

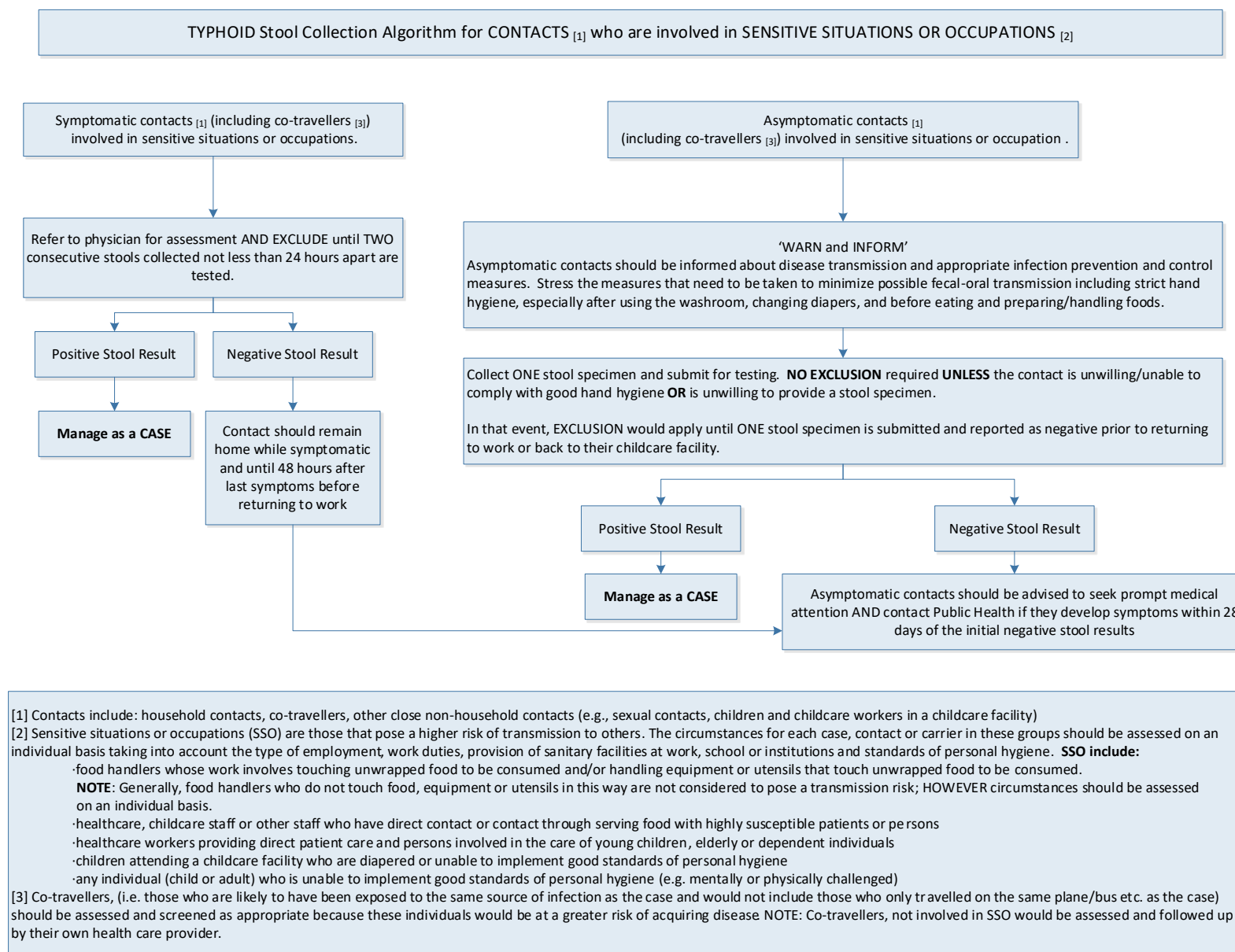
Travellers

- Encourage all travellers to:
 - seek travel advice before visiting areas of high prevalence of disease, and
 - receive typhoid vaccine before travelling to countries where there is an increased risk of exposure to *S. typhi*.
- Emphasize the importance of good sanitation, proper arrangements for safe water supplies and scrupulous personal hygiene while travelling.

Appendix 1: Cases Involved in Sensitive Situations and Occupations



Appendix 2: Contacts Involved in Sensitive Situations or Occupations



Appendix 3: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Treatment section moved to Epidemiology • Updated web links • Added link to AIP
	Public Health Management	• Terminology changed from Sensitive Occupations or Situations to Sensitive Situations or Occupations to align with other enteric diseases • Layout and minor wording changes to improve clarity

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